

Causal Inference

Difference-in-Differences

Conceptual Introduction & Storytelling

"What is the causal effect of X on Y ?" — the central question in economics

Today's Journey

01

The Causal Question

What economists mean by causal effects — and the fundamental problem

02

London, 1854 — John Snow

The original natural experiment and the birth of causal thinking

03

The 2×2 DiD Table

From story to structure: turning intuition into a number

04

The Parallel Trends Assumption

The one thing DiD cannot give you for free — and when it fails

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Modern Application

Card & Krueger (1994): minimum wages and employment

The Central Question in Economics

What is the causal effect of X on Y?

Examples:

What is the effect of education on wages?

What is the effect of a minimum wage increase on employment?

What is the effect of pollution exposure on health?

What is the effect of a new policy on economic outcomes?

Correlation Is Not Causation

We observe: **Education** ↑ → **Wages** ↑ Does education *cause* higher wages?

Maybe not — other factors could explain both:

Ability:

Talented people pursue more education AND earn higher wages — a common cause

Family background:

Wealthier families invest in education AND provide better job opportunities

Motivation:

Driven individuals choose more schooling AND work harder at their jobs

Causal inference is about ruling out these alternative explanations — systematically.

The Fundamental Problem of Causal Inference

For each unit (person, city, firm), two potential outcomes exist:



Causal Effect = $Y(1) - Y(0)$ | The fundamental problem: we can only ever observe ONE of these.

We need a way to estimate $Y(0)$ — the outcome that never happened.

The Ideal Solution: Randomized Experiments

Randomly assign units to treatment or control — then compare outcomes.

Why randomization solves the problem:

- On average, the control group is statistically identical to the treated group — same observable and unobservable characteristics
- The control group is a valid $Y(0)$: we can directly observe what would have happened without treatment
- Treatment is independent of all other factors: $\text{Treatment} \perp \text{Confounders}$

In economics, randomized experiments are often impossible, unethical, or prohibitively expensive. Economists rely instead on natural experiments and quasi-experimental designs.

The Alternative: Natural Experiments

Situations where treatment is assigned 'as-if' randomly due to external circumstances.

Policy changes:

A law applies to some states but not others, or changes at a specific date

Geographic variation:

A river, border, or physical feature creates different treatment on each side

Historical accidents:

Timing of events creates groups that received different treatments by chance

Institutional rules:

Cutoff rules, lotteries, or quotas that determine who is treated

One of the most famous early natural experiments comes from London, 1854.

PART 02

London, 1854

A physician, a pump, and the birth of causal thinking

"Before John Snow, the scientific consensus blamed cholera on bad air. Snow believed the answer was in the data — and he was right."

London, Summer 1854: A City Under Siege

The dominant theory: Miasma — bad air from filth caused disease. Scientific consensus.

Dr. John Snow was a skeptic. He suspected **contaminated water** — but he needed evidence.

The Soho outbreak of 1854 gave him a natural experiment. Hundreds died within days — but the deaths were not evenly distributed across the neighborhood.

500+

Deaths in 10 days

1854

Year of outbreak

Soho

Epicenter, London

Key question Snow asked: Why do some households have many deaths while nearby neighbors have none?

The Natural Experiment: Two Water Companies

Snow's structural observation: two companies served interleaved households — same streets, same air, same 'miasma.'

Southwark & Vauxhall Co.

Source: Thames at Hungerford
(downstream of sewage outflow)

Contamination: High — sewage in water supply

Deaths (1854): 315 per 10,000 households

Lambeth Water Co.

Source: Thames at Thames Ditton
(upstream, above sewage)

Contamination: Low — clean water source

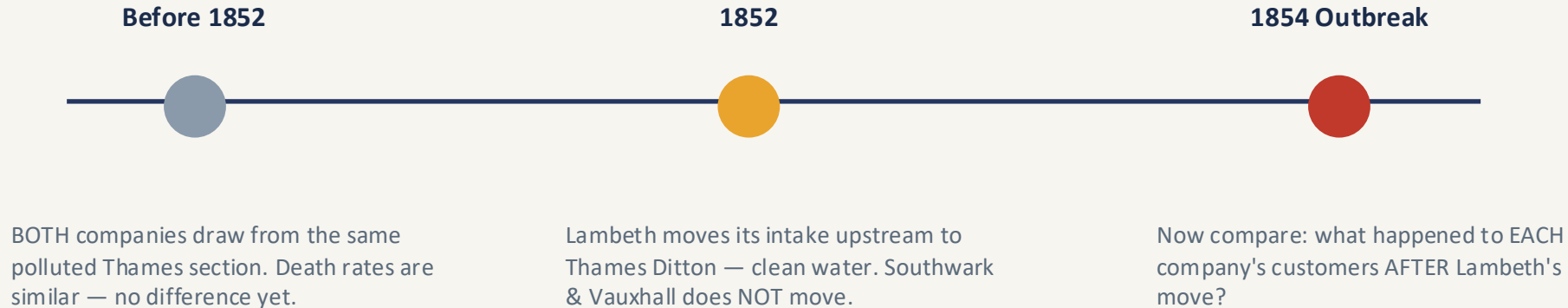
Deaths (1854): 37 per 10,000 households

The genius: The ONLY systematic difference between interleaved households was the water company that served them.

This is causal identification: treatment variation is not caused by the outcome.

Snow's Critical Insight: Add the Time Dimension

The real power came from a company decision that happened BEFORE the 1854 outbreak.



DiD structure: Two groups (Lambeth = treated, S&V = control) × Two periods (Before 1852, After 1852) = the 2×2 table.

From John Snow to Modern Econometrics

What Snow Did (1854)

Identified two groups: polluted vs. clean water

Observed outcomes BEFORE Lambeth's 1852 move

Observed outcomes AFTER the move (1854 outbreak)

Used the unchanged group (S&V) as the counterfactual

What DiD Does (today)

Identify a treated group and a control group

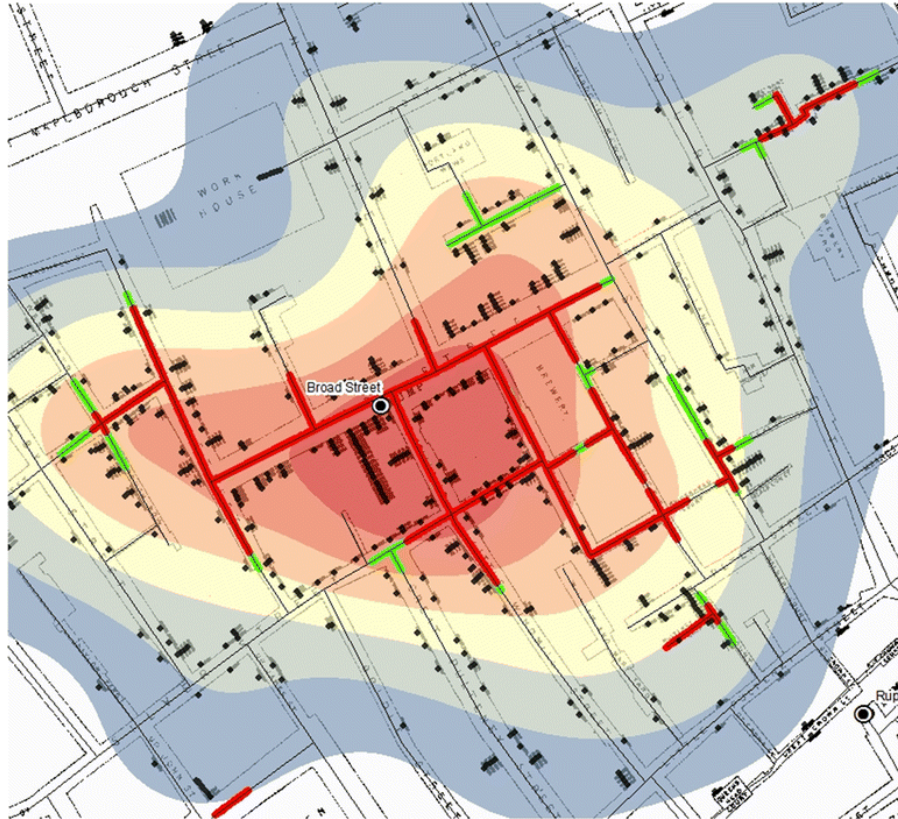
Collect outcome data BEFORE the policy change

Collect outcome data AFTER the policy change

Use the control group's trend to build the counterfactual

Difference-in-Differences formalizes Snow's logic into a rigorous econometric estimator.

Map of the Cholera Outbreak



PART 03

The 2×2 DiD Table

Turning intuition into a number

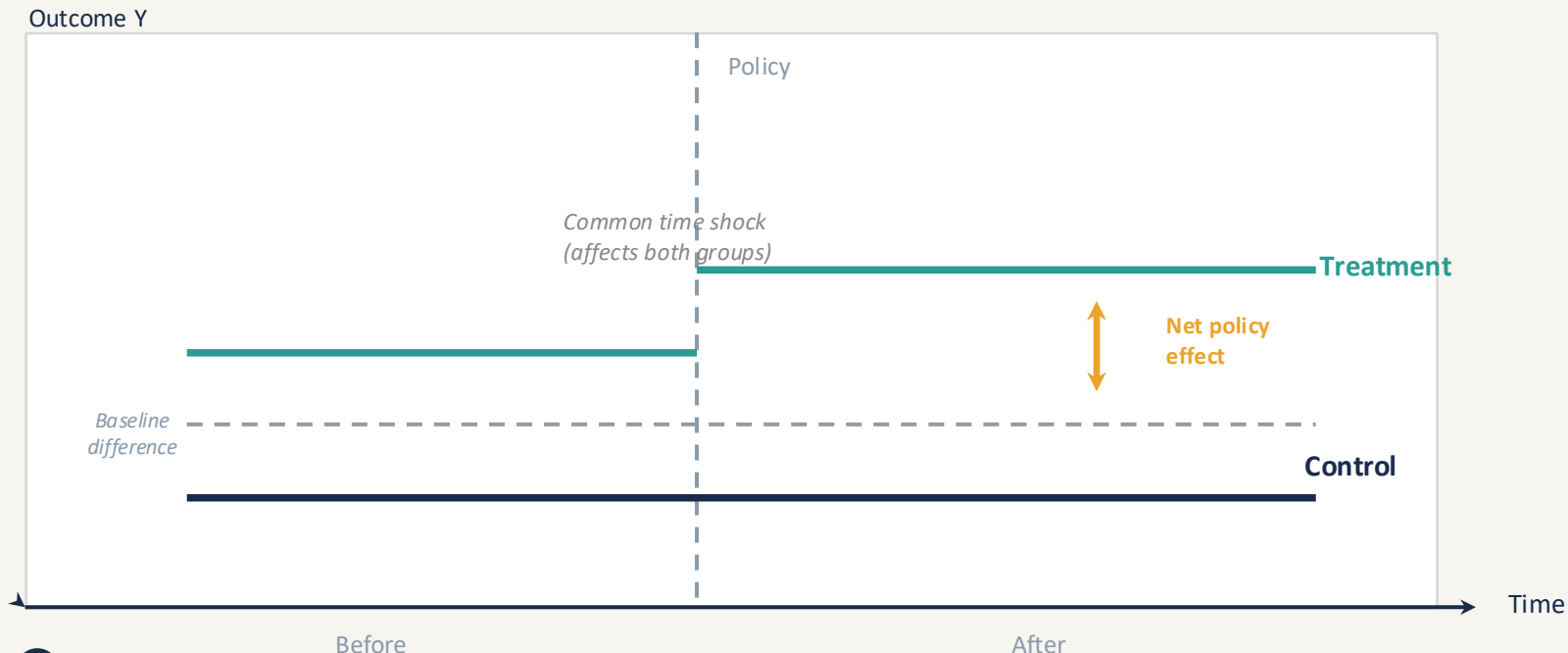
The Difference-in-Differences Table

	Before (1849)	After (1854)	Difference
Lambeth (Treated)	130	37	-93 $\leftarrow \Delta T$
S&V (Control)	135	315	+180 $\leftarrow \Delta C$
DiD (T - C)	-5 ≈ 0	-278	-273 $\leftarrow \text{DiD}$

(Deaths per 10,000 households)

DiD = (37 - 130) - (315 - 135) = -93 - (+180) = -273 Lambeth's cleaner water saved ≈ 273 lives per 10,000.

Why We Subtract Twice: What DiD Removes



- First difference removes baseline level differences between groups (treated vs. control)
- Second difference removes common time shocks that affect both groups equally (before vs. after)
- What remains — under parallel trends — is the net causal effect of the policy

PART 04

The Parallel Trends Assumption

The one thing DiD cannot give you for free

"DiD is only as good as the assumption that, in the absence of treatment, both groups would have followed the same trend."

The Parallel Trends Assumption

Formal Statement:

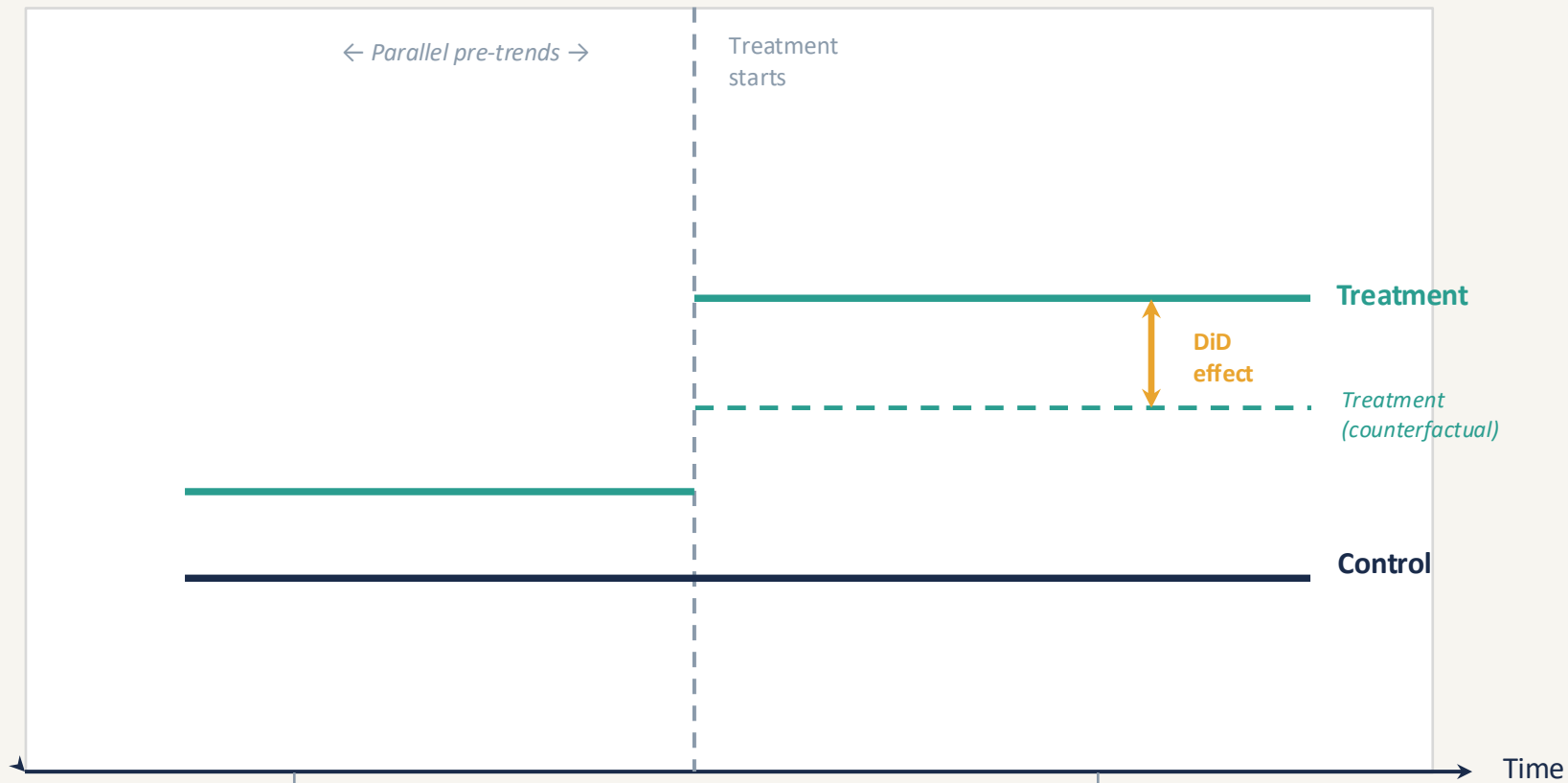
In the absence of treatment, the average outcome for the treated group would have changed over time by the same amount as the average outcome for the control group.

In plain English:

- ✓ Groups can start at DIFFERENT LEVELS — parallel trends does NOT require identical baseline outcomes
- ✓ Groups must have the SAME TREND before the treatment — the slopes must be parallel in the pre-period
- ✓ This trend must be what WOULD HAVE CONTINUED after treatment if nothing had changed
- ! This is an assumption about the counterfactual — it can never be directly tested, only made plausible

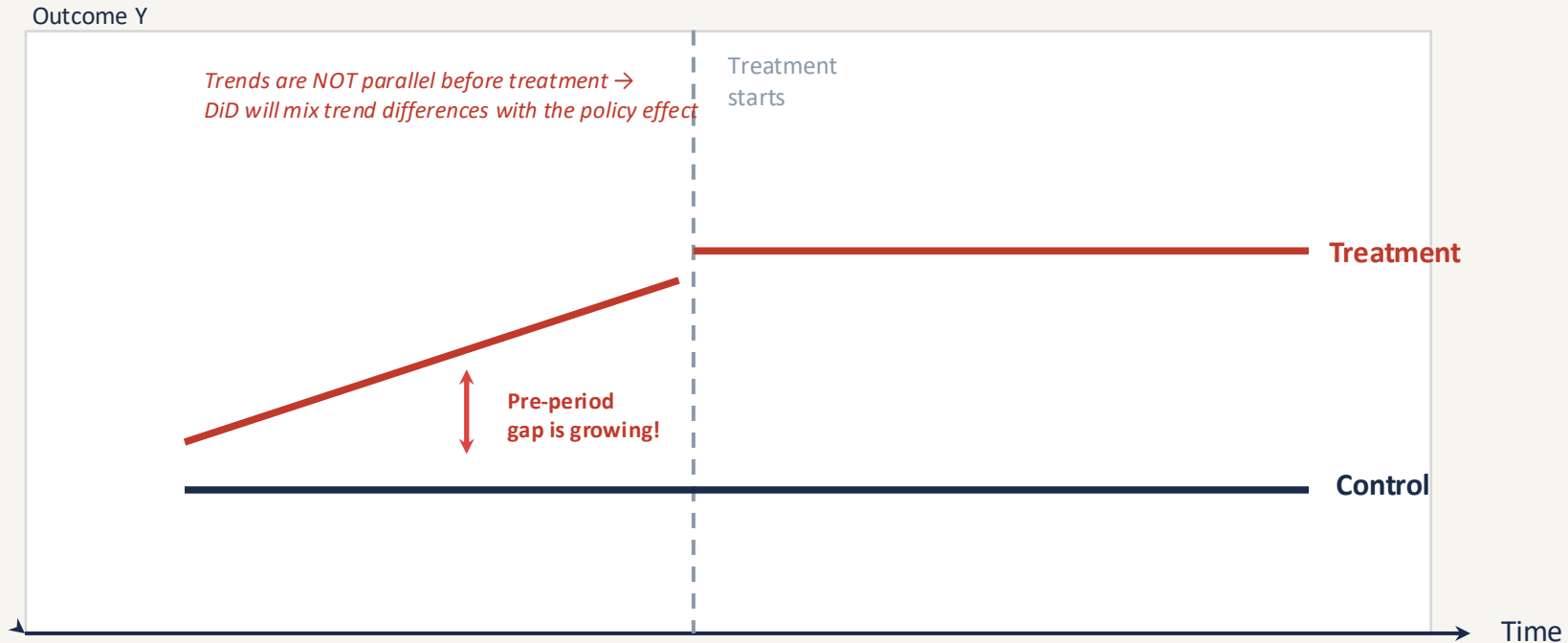
DiD Graphically: The Clean Case

Outcome Y



The control group's trend builds the counterfactual for the treated group. DiD = gap between actual and counterfactual.

When DiD Fails: Non-Parallel Pre-Trends



- **Selection bias:** The treated group was already on a different trajectory before any treatment
- **Simultaneous shocks:** Something else changed at the same time, but only for the treated group
- **Composition changes:** The type of units in each group changes between periods (e.g., people move)

Was Parallel Trends Plausible? Back to Snow.

Every DiD paper must evaluate this assumption explicitly — let's practice with Snow.



Same neighborhoods

Lambeth and S&V customers lived on the same streets — same air, poverty, miasma. Only the water source differed.



Similar pre-treatment trends

Before 1852, BOTH companies drew from the same polluted Thames section. Death rates were comparable — pre-trends were parallel.



Exogenous treatment assignment

Lambeth moved its intake for business/engineering reasons — not because its customers were already getting healthier. No selection bias.



Potential concern: wealth differences

Wealthier households may have been more likely to be served by one company. Snow addressed this — the effect held even controlling for wealth.

Modern Application: Card & Krueger (1994)

What is the effect of raising the minimum wage on employment?

A natural experiment from a state border:

New Jersey — TREATED

Minimum wage raised from \$4.25 → \$5.05
Fast-food employment observed before and after

Pennsylvania — CONTROL

Minimum wage did not change
Same fast-food restaurants observed over same period

The finding:

Employment in New Jersey's fast-food sector did NOT fall after the minimum wage increase. Using Pennsylvania's employment trend as the counterfactual, Card & Krueger found the wage hike had no significant negative effect — directly challenging the textbook prediction and reshaping policy debates.

What We Built Today

01

Causal effects require counterfactuals

We want $Y(1) - Y(0)$, but can only observe one. A valid comparison group estimates the missing potential outcome.

02

John Snow showed us the way

Two water companies, same streets, one policy change — a natural experiment born from circumstance.

03

DiD = subtract twice

First diff: change within each group over time. Second diff: subtract the control's change to remove common shocks.

04

Parallel trends is the key assumption

Without it, DiD mixes trend differences with the policy effect. Always check and defend it explicitly.

05

Real applications are everywhere

Card & Krueger (1994) used exactly this logic to challenge the textbook prediction on minimum wages.

Next: Implementing DiD in Python — regression form + testing parallel trends with real data